

GAME DESIGN DOCUMENT

GridMancer

“Arrange your arsenal and survive the dungeon”

1. Game Overview

1.1. Game Summary

Title: GridMancer
Genre: 2D Tactical Puzzle
Platform: PC (Windows)
Game Jam: Compfest Game Jam 2026
Theme: Order in Disorder
Development Team: 3 person
Development Time: 3 days
Engine: Unity

GridMancer is a 2D tactical puzzle game where players strategically place unique weapons on a limited grid to defeat enemies. Each weapon offers distinct properties and can interact with others to create powerful synergies, but once placed, it cannot be moved, making every placement and attack order a critical decision. As levels become increasingly complex, players must organize limited options, optimize their weapon combinations, and adapt their strategy while racing against the clock.

1.2. Design Motivation

I've always been interested in how the Resident Evil inventory system implements its item-matching puzzle mechanics. This inspired me to explore how similar mechanics could be applied more directly to combat gameplay.

Inspired by the recent game Overgeared Hero, GridMancer takes item-matching mechanics and applies them to a more direct combat system. Every placement is permanent, and each item has unique synergies with others. Players must therefore carefully strategize their weapon placement on the grid.

1.3. Theme Integration

The theme "Order in Disorder" is reflected through the game's permanent grid-placement system. Players encounter varied grid layouts alongside a limited and seemingly disorganized selection of weapons, challenging them to create order through careful placement and attack sequencing. Since placements cannot be changed, every decision carries lasting consequences, requiring players to transform limited resources and space into an organized strategy capable of defeating each enemy.

1.4. Design Pillars

The game is designed around five primary pillars:

1. Strategic Placement

Every item placement is permanent, so players must think carefully about where each weapon belongs. The grid should encourage players to plan ahead, consider future placements, and make meaningful decisions rather than constantly rearranging their inventory.

2. Synergy

Items should become more powerful when combined with compatible weapons. Players are encouraged to discover relationships between items and build combinations that complement each other.

3. Resource Management

Players have a limited number of weapons, grid spaces, and time. Every placement consumes permanent grid space, forcing players to weigh immediate damage against the potential value of future placements.

4. Adaptation

Players should constantly adapt their strategy based on the items they acquire and the challenges they encounter. There should rarely be a single perfect arrangement; instead, players are encouraged to experiment with different placements and synergies to respond to changing combat situations.

5. Order in Disorder

The grid represents the tension between organization and chaos. Although enemies and combat can quickly become unpredictable, players must create order through careful item placement.

2. Gameplay

2.1. Gameplay Flow

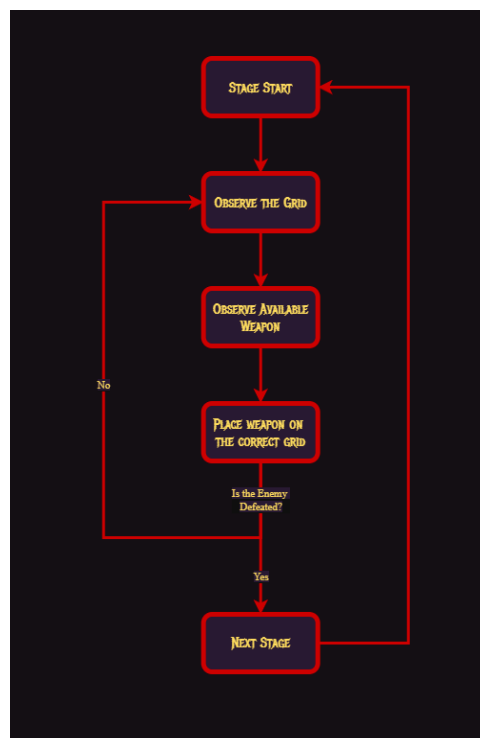
In GridMancer, players progress through a dungeon by defeating enemies using a combination of strategic planning and weapon placement.

Each encounter presents the player with a unique grid layout and a selection of weapons with different effects and properties, requiring them to carefully arrange and sequence their weapons to create effective attack combinations and maximize damage. The challenge lies in identifying the optimal placement and weapon order for each encounter, as every decision can determine the outcome. As players progress through the dungeon, grid layouts become increasingly complex and enemies more challenging, forcing them to continuously adapt their strategies and make increasingly deliberate decisions.

The player must defeat the enemy by depleting its health to 0 before running out of available weapons, grid space, or time. Successfully reducing the enemy's health to 0 completes the encounter and allows the player to progress to the next stage. If the player uses all available weapons without defeating the enemy, fills all available grid spaces while the enemy is still alive, or allows the timer to reach zero, the enemy attacks and defeats the player. The player must then restart from the beginning.

2.2. Core Gameplay Loop

The primary gameplay loop consists of:



3. Mechanics

GridMancer features several mechanics:

3.1. Grid

The Grid is the main play area where players plan and execute their attacks. Each stage features a predetermined grid layout, with different stages introducing different arrangements. As the player progresses through the dungeon, the grid layouts become increasingly complex, requiring players to carefully consider how they use the available space.

3.2. Weapon

Weapons are the player's primary method of dealing damage to enemies. Each stage provides the player with a variety of weapons, each with its own properties, quirks, and potential synergies with other weapons.

	🔍 Weapon Name	🕒 Stage	🔍 Damage Type	⚖️ Base Damage	⚡ Special Effect	💰 Resource Cost
1	Sword	Stage 1	Direct Damage	3	-	2 horizontal
2	Staff	Stage 1	Direct Damage	2	-	2 vertical
3	Poison Dagger	Stage 2	Hybrid	1	Applies poison dealing 1 damage forever	1
4	Book of Multiplier	Stage 1	Buff	0	Multiply the base damage of staff	1
5	Book of Addition	Stage 1	Buff	0	Add damage on top of base damage on staff	1
6	Hourglass	Stage 2	Buff	0	Repeat all weapon near the hour glass	2 vertical

When a weapon is placed on an available space on the grid, it immediately deals its damage to the enemy and permanently occupies that space. Weapons cannot be moved or removed after being placed, making every placement a permanent decision. Players must therefore carefully consider both the order and position of their weapons to maximize their effectiveness.

3.3. Inventory System

At the beginning of each stage, the player is provided with a set of weapons, which are added to their inventory. When a weapon is placed during an encounter, it is permanently consumed and cannot be recovered or reused within that stage.

Inventory persists between stages, meaning that unused weapons remain available for future encounters. As the player progresses, a small number of new weapons may be added to their existing inventory.

This creates a resource-management decision that extends beyond individual encounters, as players must decide whether to use powerful weapons to secure the current stage or conserve them for future challenges. Using too many strong weapons early may leave fewer effective options for later stages, while conserving them can make the current encounter more difficult.

The system encourages players to balance immediate survival with long-term resource management, adapting their strategy based on the weapons available to them and the challenges ahead.

3.4. Enemy

Enemies are the primary obstacles that players must defeat to progress through the dungeon. Each enemy has a set amount of health, which the player must reduce to zero using their available weapons.

There are 2 types of enemies

- **Basic Enemies**

Basic Enemies have a fixed amount of health and do not possess any special abilities. They provide straightforward encounters that primarily test the player's ability to utilize their weapons and available grid space effectively.

- **Boss Enemies**

Boss Enemies are more complex encounters with two distinct phases. Like Basic Enemies, they have a set amount of health, but reaching a specific health threshold causes the Boss to enter its second phase. During the second phase, the Boss gains special properties or mechanics that change how the player must approach the encounter.

3.5. Timer

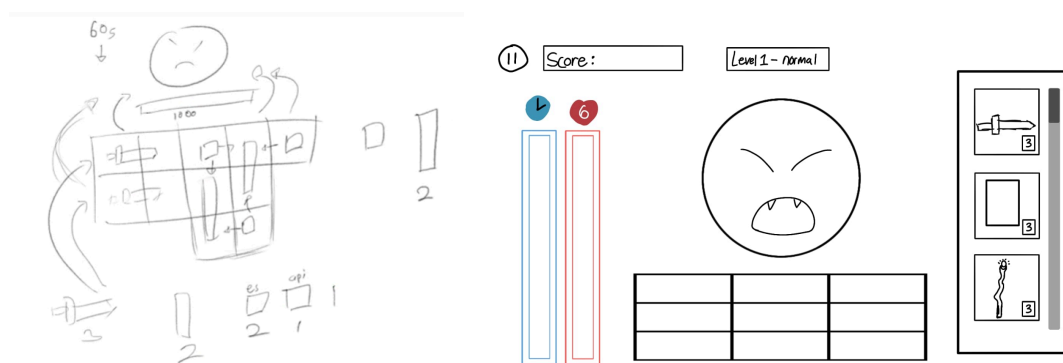
Each stage has a 60-second timer that limits the amount of time the player has to complete the encounter. Within this time, players must analyze the grid, determine the most effective weapon order and placement, and defeat the enemy. The timer continuously counts down throughout the encounter, creating pressure for players to

balance careful planning with quick decision-making. If the timer reaches zero before the enemy is defeated, the player loses the encounter and must restart from the beginning.

4. Stage and Progression

4.1. Stage System

Each stage consists of a predetermined grid layout, a set of weapons, and an enemy encounter. The enemy appears with a set amount of health, while a 60-second timer continuously counts down throughout the stage. The player must carefully arrange and use their available weapons on the grid to deal enough damage and defeat the enemy before the timer runs out. Successfully defeating the enemy allows the player to progress to the next stage.



4.2. Stage Progression

GridMancer consists of three stages that gradually introduce and test the game's core mechanics. The first two stages feature Basic Enemies and are designed to familiarize players with arranging their limited weapons within the available grid space. As the player progresses, the grid layouts and weapon requirements become more challenging, requiring greater consideration of weapon placement and sequencing.

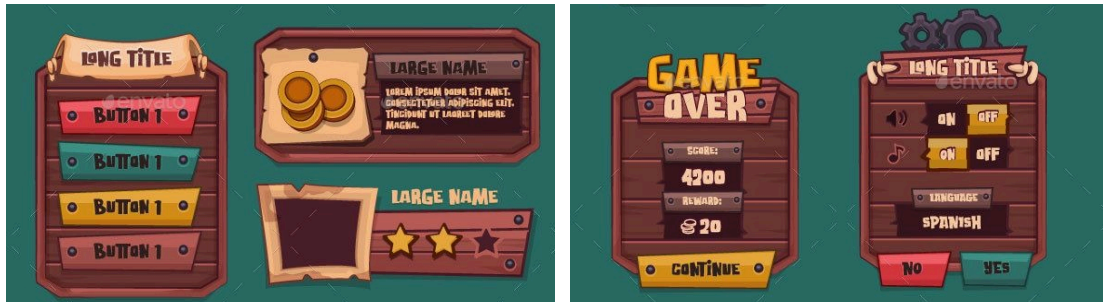
The third stage is a Boss encounter consisting of two phases. During the first phase, the Boss behaves according to its initial properties. Once the Boss enters its second phase, it becomes immune to poison damage. This forces the player to adapt their strategy and rely on weapons that deal direct damage instead of poison.

After successfully defeating the Boss, the player completes the dungeon and reaches the end of the game.

5. UI / UX

5.1. UI/UX Direction

The UI/UX direction of GridMancer takes inspiration from dungeon-themed RPGs and medieval fantasy settings. The UI is designed to feel approachable and fun while still supporting the game's challenging and strategic gameplay.



5.2. Main Menu

The Main Menu features a medieval-themed background that establishes the game's dungeon atmosphere. The interface is kept simple and easy to navigate, with illustrated titles and buttons that contrast against the darker and colder environment. The combination of stylized illustrations, textures, and typography helps establish GridMancer's playful medieval identity.

5.3. Gameplay

During gameplay, every important element remains visually prominent and easy to understand. The grid is positioned at the center of the scene, with the available weapons, timer, and enemy health displayed around it for quick access and visibility. When a weapon occupies a grid cell, the cell changes color to clearly indicate that it is occupied, and the weapon becomes permanently fixed in place.

Weapons display tooltips when hovered, providing a quick overview of their properties, quirks, and potential synergies. A counter shows the number of available weapons, with a +1 indicator appearing whenever a new weapon is added at the start of a stage.

The timer and enemy health are represented as vertical bars for clear feedback. The timer gradually decreases as time passes, and when time is running low, the bar pulses red to communicate urgency and encourage quick decision-making. Enemy health is accompanied by a numerical value showing the remaining health.

During boss fights, introductory text appears to signal the start of the encounter. When the boss enters Phase 2, a similar message introduces the new phase and explains its mechanics. The introductory message disappears after a few seconds, while the mechanic explanation remains visible so players can refer back to it if they miss or forget the information.

5.4. Tutorial

The tutorial uses a simple and straightforward interface in the form of a book, to introduce the game's mechanics. It explains each weapon's properties, quirks, and potential synergies, allowing players to understand how different weapons interact before applying them during gameplay.

6. Art and Visual

6.1. Art Style Overview

GridMancer uses a rough, cartoon-like art style inspired by medieval-themed RPGs and dungeon environments. The visual style combines vibrant colors, exaggerated shapes, and rough textures to create a playful yet grungy atmosphere. This contrast is intended to make the game feel fun and approachable while still communicating a sense of danger and challenge.

The overall visual identity is characterized by:

- Vibrant colors
- Rough and weathered textures
- Strong contrast in values
- Vibrant yet grungy atmosphere

The overall art direction should maintain a balance between playful and threatening, ensuring that the visual style supports both the game's strategic gameplay and its medieval dungeon setting.

6.2. Art Direction

The art direction aims to achieve the following:

6.2.1 Readability

The player's attention should be immediately drawn to the grid, weapons, and timer, making the main gameplay elements easy to recognize and understand at a glance.

6.2.2. Playful Yet Dangerous

The game should have a playful and inviting appearance, while still conveying a sense of danger and tactical gameplay through its visual elements.

6.2.3 Consistency

Enemies, objects, backgrounds, and UI should follow a consistent rough, cartoon-like medieval style, ensuring that all visual elements feel cohesive within the game.

6.3. Enemy Art

Enemies feature a rough, cartoon-inspired aesthetic heavily influenced by classic fantasy dungeon monsters. Their designs should communicate a strong sense of danger and tension while retaining a vibrant, playful, and visually enticing appearance. Exaggerated silhouettes, expressive features, rugged textures, and bold shapes can reinforce their threatening nature without losing the charm of the overall art style.

6.4. Environment Art

Environments should feature a dark, desaturated color palette to create strong contrast against the vibrant objects and enemies. They should maintain the same medieval-inspired aesthetic, with dungeon environments serving as the primary backdrop throughout every stage. Dim lighting, aged stone, worn surfaces, and atmospheric details should reinforce the dungeon setting while allowing gameplay elements to remain visually prominent and easy to distinguish.

6.5. Shading and Lighting

The art should use soft, stylized shading with a strong emphasis on textured surfaces. Rather than relying on sharp or highly realistic lighting, the shading should enhance the rough and grungy qualities of the visuals. Textures, subtle gradients, and atmospheric lighting should add depth and reinforce a sense of danger, creating contrast with the otherwise vibrant color palette while maintaining the overall cartoon-like aesthetic.

7. Music and Audio

7.1. Audio Direction

The audio design aims to create a balance between fun and tension, complementing the game's vibrant visual style while reinforcing its dangerous dungeon atmosphere. The music should maintain a fast-paced and tense feel inspired by RPG soundtracks, while sound effects should provide clear and satisfying feedback for every player interaction and attack.

7.2. Music

The game features two primary music tracks, both generated using Suno AI.

The menu music uses a slow-paced techno composition to establish a calm yet playful atmosphere and encourage the player to begin the game. Its relaxed rhythm contrasts with the menu's dark visual background, creating an engaging balance between mood and presentation.

The gameplay music takes a more intense approach, featuring a fast-paced techno composition with heavy percussion and an increasing sense of tension and urgency. The energetic rhythm reinforces the time-sensitive nature of gameplay, encouraging players to think and strategize quickly before time runs out.

7.3. Sound Effects

All sound effects are sourced from Pixabay and are used to provide immediate audio feedback for player actions, making interactions feel more responsive and immersive. Each attack and interaction is assigned a distinct sound effect to improve clarity, reinforce the player's actions, and enhance the overall sense of immersion.

Gameplay sound effects include:

- Placing weapons on the board
- Weapon attacks
- Enemy damage and death
- Boss fight and Phase 2 introduction
- UI interactions, such as buttons and menus

8. Development Process

8.1. Team

Our team is composed of 3 people, each with their own role and tasks.

Member	Role
Adriel Kevin Jonathan	Game Designer, Programmer, QA
Darryl Arief Tananjaya	Game Designer, Programmer
Jocelyn Tania Harjanto	2D Artist

8.2. Schedule

The prototype was developed for Compfest Game Jam 2026, with a strict development period of three days, from 2 to 4 August 2026. To effectively plan and manage the limited development time, the team used Miro as a collaborative planning tool. It was used to brainstorm, organize, and filter ideas, establish development priorities, define each team member's tasks, and create the overall development schedule.

	Task Name	Developer	Status	Notes	Dependencies
1	Drag n Drop	Adriel	Complete	Import Ecorangers	-
2	Timer	Adriel	Complete	Import Ecorangers	-
3	Grid system (detection and snapping)	Adriel	Complete	Handle weapon grid size detection and snapping adjustment	Weapon System
4	Health and Enemy	Darryl	Complete	Handle enemy health and state change	Grid System and Weapon System
5	Weapon system	Darryl	Complete	Handle weapon base damage, grid size, and relation	-
6	Inventory System	Darryl	Complete	Handle weapon count	Weapon System
7	Weapon Effects and Buffs	Darryl	Complete	Handle weapon modifier (multiplier, addition, bonus attack)	Weapon System
8	Win and Lose	Adriel	Complete	Handle endgame state	Health and Enemy
9	Stage system	Adriel	Complete	Handle stage transition and data persistence	Win and Lose
10	UI system	Adriel	Complete	UI :) (menu and game over)	-
11	Handbook	Darryl	Complete	Design and explanation	-
12	Music and SFX	Darryl	Complete	Audio Source and Manager	-
13	VFX and Juice	Darryl	Complete	Feedback and visual effects	-
14	Level Design	Darryl & Adriel	Complete	Final Polishing	All

Weeks ▾ 📅 🗨

August

14 records

Theme drops and brainstorm

Pick idea with clearest core mechanic

Build main interaction loop

Set up project structure and basic UI

Sketch art style and enemy concept

Deliver enemies, environment tiles, and key pr

Build levels and content pipeline

Build menus, tutorial, win/lose states

Deliver remaining sprites, backgrounds, UI

Internal playtesting and bug fixes

Final art pass and audio balance

Difficulty tuning

Build for all platforms and write description

Submit game

2 Satisfying Mechanic

Theme drops and brainstorm

Pick idea with clearest core mechanic

Build main interaction loop

Set up project structure and basic UI

Sketch art style and enemy concept

3

Complete Game

Deliver enemies, environment tiles, and key props

Build levels and content pipeline

Build menus, tutorial, win/lose states

Deliver remaining sprites, backgrounds, UI

4

Submitted

Internal playtesting and bug fixes

Final art pass and audio balance

Difficulty tuning

Build for all platforms and write description

Submit game

9. Development Notes

9.1. Scope

GridMancer is a game developed within three days for Compfest Game Jam 2026. Given the limited development time, the game's scope was intentionally restricted to a small set of simple mechanics and a limited number of stages. This approach allowed the team to prioritize and refine the core gameplay experience while minimizing the risk of scope creep.

9.2. Constraints and Challenges

The game was developed around the theme "Order in Disorder." Combined with the strict three-day development period, this theme presented significant constraints on the team's creative process. As game developers, we needed to design a game that not only fit the theme, but was also achievable within the limited timeframe, while remaining simple and, most importantly, fun to play.

Throughout the development process, the team encountered several challenges:

- Developing a creative and unpredictable game concept that effectively fits the given theme.
- Dividing tasks and allocating time efficiently within the limited three-day development period.
- Creating a responsive grid-based UI and implementing a functional inventory system.
- Maintaining effective communication and coordination between team members throughout development.

9.3. Limitations

Due to the short development timeframe and limited team resources, the final version of GridMancer contains three stages and six weapon variations, with a limited number of weapon synergies. While the team initially planned to include additional weapon types and more complex synergies, these features had to be reduced in scope to prioritize the six core weapons and ensure that the main gameplay experience could be completed within the jam's timeframe. Additional stages were also planned but were not implemented due to the available development time.

9.4. Lessons Learned

Developing GridMancer within three days taught the team that effective planning and communication are essential to maintaining a smooth development process. Our initial schedule did not fully align with the actual development process, which led to early miscommunication regarding ideas, priorities, and task assignments.

Communication and scheduling also presented challenges during development, as team members had to balance the project with their individual responsibilities. This occasionally made it difficult to coordinate tasks and synchronize progress within the limited development timeframe. As a result, some planned aspects of the game had to be adjusted. For example, due to the limited time available, the team decided to use AI-generated music rather than spending additional time searching for suitable soundtracks.

This experience gave the team a clearer understanding of the importance of realistic scheduling, task prioritization, and consistent communication. In future projects, the team can improve its workflow by allocating time more carefully, establishing clearer task ownership, and regularly communicating progress, concerns, and potential roadblocks. Identifying these issues earlier would allow the team to adapt more effectively without significantly disrupting the development process.